Bayesian Hyperspectral Image Super-Resolution in the Presence of Spectral Variability

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***Abstract*— Synthesizing a high-resolution (HR) hyperspectral image (HSI) by merging a low-resolution (LR) HSI with a corresponding HR multispectral image (MSI) has become a promising HSI super-resolution scheme. Most existing HSI–MSI fusion methods are effective to some extent, while several challenges remain. First, the spectral response of a given material exhibits considerable variability due to different acquisition times and conditions, however, variations in spectral signatures are often neglected. Second, a majority of off-the-shelf methods require predefined degradation operators, which can be unavailable in practice. To tackle the above issues, we introduce a novel fusion approach with a Bayesian framework. Specifically, we regard the up-sampled LR-HSI as the low-frequency component of the underlying HR-HSI. We characterize the texture features of high- and low-frequency components, respectively, which can enlarge modeling capacity and bypass the absence of degradation operators. Furthermore, we depict the relative smoothness of reflectance spectra with the Gaussian process. Extensive experiments on synthesized and real datasets illustrate the superiority of the proposed strategy in terms of fusion performance and robustness to spectral variability.**

***Index Terms*— Bayesian inference, hyperspectral image (HSI), image fusion, multispectral image (MSI), spectral variability, super-resolution.**

I. Introduction

HYPERSPECTRAL imaging system can preserve electromagnetic spectra of the detected area at hundreds of wavelength intervals. The spectral details of hyperspectral images (HSIs) enable a myriad of applications requiring fine identification of materials, such as classification [1], target detection [2], and so on. However, HSIs severely lack in terms of spatial resolution due to the restriction of incident energy [3], which hinders the widespread applications of HSIs. Therefore, increasing efforts have been invested in HSI super-resolution.

The ultimate goal of HSI super-resolution is to improve the spatial resolution of HSIs. Nevertheless, increasing the spatial resolution of hyperspectral sensors is straight to achieve. Over the past decade, there has been a renewed interest in HSI super-resolution by fusing HSIs with corresponding higher-resolution multispectral images (MSIs) [4], which has been proven to be a more compact and efficient HSI super-resolution scheme.

In the HSI–MSI fusion issue, the observed images are depicted as degradations of the underlying high-resolution (HR) HSIs, and therefore, the reconstruction of HR-HSI can be cast into a variational problem. A variety of HSI-MSI fusion approaches have been flourishingly developed in recent decades. Typically, representative HSI–MSI fusion techniques can be roughly categorized as model optimization-based, Bayesian-based, and deep learning-based methods.

Motivated by the advances in modern optimization theory [5], [6], model optimization has become one of the most commonly applied HSI–MSI fusion techniques. Given the fact that HSI resides in a lower-dimensional subspace spanned by spectral signatures [7], [8], the unmixing/subspace representation-based methods characterize the HR-HSI as the linear combination of several basis vectors [9], [10], [11], [12], [13], [14], [15]. More recently, tensor decomposition [16], [17] has become a prevalent tool for HSI–MSI fusion [18], [19], [20], [21], [22], which seeks feature mapping to project the high-order HSI into a succinct representation without destroying intrinsic correlations.

Guided by Bayesian theory [23], [24], HSI–MSI fusion is recast as a maximum a posteriori issue, where the objective function is based on the prior knowledge and degradation models. Bayesian sparse representation (BSR) [25] is a representative Bayesian nonparametric HSI–MSI fusion method, which explored the sparsity of HR-HSI. Fast fusion based on Sylvester equation [26] exploited the property of circulant matrix and solved the underlying Sylvester equation directly.

With the development of deep learning [27], deep neural network (DNN) has got involved in the HSI analysis [28]. In [29], an iterative HSI–MSI fusion approach based on a deep HSI denoiser was introduced to leverage both domain knowledge likelihood and deep image prior. The unsupervised

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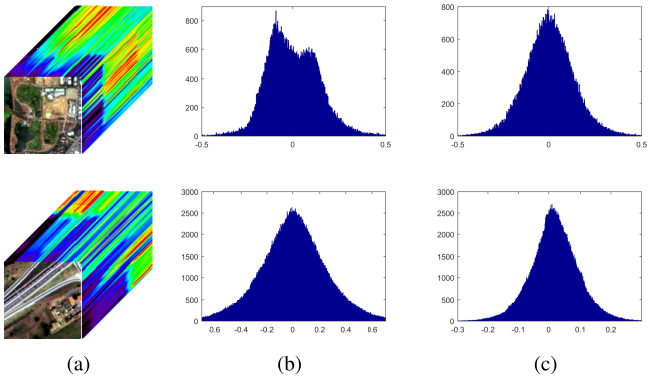
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adaptation learning [30] designed a two-stage DNN, namely, a fusion module and an adaptation module, which can recover the latent HR-HSI in a coarse-to-fine scheme. Zhang et al. [31] designed an image-specific generator to produce HR-HSI and structured degradations in spectral and spatial through a fully connected layer and convolutional layer, respectively. Dian et al. introduced a Convolutional Neural Network denoiser [32] based FUSion framework (CNN-FUS) [33] for HSI super-resolution.

Though HSI–MSI fusion provides a great opportunity for HSI super-resolution, several challenges may arise in practical applications. First, the observed images are often captured at different time instants, leading to significant variability in spectral signatures caused by varying illumination, atmospheric distortion, and so on. Second, the degeneration processes in the spectral and spatial domains are assumed to be predefined or can be estimated. While this assumption cannot be satisfied in practical applications since the spatial degradation may involve several unknown factors such as blurring and down-sampling operators [18], [34]. These issues are often neglected in HSI super-resolution approaches proposed to date and tend to have significant influences on the effectiveness of practical applications.

To simultaneously address the above problems, we introduce a fully Bayesian generative model, which has gained satisfactory performance in terms of HSI super-resolution and robustness to spectral variability. Motivated by the HSI-MSI fusion with spectral variability (FuVar) [35], we propose to model variations of reflectance spectra with positive scaling factors. We emphasize that our method essentially differs from [35] in that the proposed model is guided by Bayesian theory and can data-adaptively fit the complex spectral variability without tuning parameters. To circumvent the absence of point spread function (PSF), we up-sample the low-resolution (LR) HSI and treat the resampled one as the low-frequency component of HR-HSI. Similar to previous work [36], we characterize the high- and low-frequency components, respectively, while our model assumptions are more reasonable. As shown in Fig. 1, the texture features of high-frequency components may exhibit complex distributions. The Gaussian distribution is significantly limited when it comes to modeling real HSIs, whereas the combination of Gaussians can give rise to complex densities [37]. Therefore, we use a mixture of Gaussians for high-frequency texture characterization. Our method has several significant characteristics, which can be summarized as follows.

1) We introduce a novel HSI super-resolution model that accounts for influences of spectral mismatches between fusion materials and exhibits robustness to variations in spectral signatures.
2) A potential spatially varying degeneration operator can be introduced into our model without sacrifice in fusion performance, as the spatial-invariant assumption is not required in our model.
3) To get rid of the parameter setting, we determine the parameters by placing noninformative hyperpriors over them. Therefore, most of the parameters can be learned adaptively during the inference process.



(a) (b) (c)

Fig. 1. Texture features of high- and low-frequency structures in two HSIs. (Top) Moffett dataset. (Bottom) Pavia dataset. (a) HSIs. (b) Texture features of high-frequency components. (c) Texture features of low-frequency components.

4) We introduce an efficient optimization scheme to derive analytical inference.

The rest of this article is structured as follows. Section II presents the formulations of HSI–MSI fusion without spectral variability and models HSI super-resolution with subspace representation. The model specifications of the proposed Bayesian HSI super-resolution in the presence of spectral variability (BSRSV) and corresponding inference algorithm are given in Section III. Comprehensive experimental results conducted on five datasets are illustrated in Section IV. The conclusion of this work is given in Section V.

II. Problem Formulation

In this work, the latent HR-HSI is denoted as $\mathbf{F} \in \mathbb{R}^{L\times MN}$, where L, M, and N correspond to the dimensions of spectral, width, and height mode, respectively. The observations HR-MSI $\mathbf{M} \in \mathbb{R}^{l\times MN}$ and LR-HSI $\mathbf{H} \in \mathbb{R}^{L\times mn}$ are regarded as spectrally and spatially degeneration products of HR-HSI. In this case, $l \ll L$, $M/m = N/n = sf$, and sf is the spatial scaling factor. Specifically, the acquisition process of $\mathbf{M}$ and $\mathbf{H}$ can be represented as

$$\begin{aligned}\mathbf{M} &= \mathbf{RF} + \mathbf{N}_m \\ \mathbf{H} &= \mathbf{FP} + \mathbf{N}_h\end{aligned} \tag{1}$$

where $\mathbf{R} \in \mathbb{R}^{l\times L}$ is the spectral response function of MSIs. The matrix $\mathbf{P} \in \mathbb{R}^{MN\times mn}$ is the spatial degeneration operator including PSF and down-sampling associated with $\mathbf{F}$. $\mathbf{N}_h$ and $\mathbf{N}_m$ are additive terms that include sensor noises and modeling errors.

The subspace representation assumes the spectral pixels of HSIs reside in a low-dimension subspace spanned by a set of basis vectors. Mathematically, the underlying $\mathbf{F}$ can be represented as

$$\mathbf{F} = \mathbf{ES}^\top \tag{2}$$

where $\mathbf{E} \in \mathbb{R}^{L\times K}$ is the dictionary containing the spectral signatures of $\mathbf{F}$ and $\mathbf{S} \in \mathbb{R}^{MN\times K}$ is corresponding texture feature. $K \ll L$ is a prefixed parameter defined by the number of basis vectors.

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The subspace representation-based HSI–MSI fusion aims at reconstructing $\mathbf{F}$ from $\mathbf{H}$ and $\mathbf{M}$ under assumption models (1). Based on the maximum likelihood estimation of $\mathbf{F}$, the HSI–MSI fusion can be reformulated as

$$\min_{\mathbf{E},\mathbf{S}} \left\|\mathbf{M}-\mathbf{R}\mathbf{E}\mathbf{S}^{\top}\right\|_F^2+\left\|\mathbf{H}-\mathbf{E}\mathbf{S}^{\top}\mathbf{P}\right\|_F^2+\lambda\varphi\left(\mathbf{E},\mathbf{S}\right) \quad (3)$$

where $\|\cdot\|_F$ is the Frobenius norm and $\varphi(\mathbf{E},\mathbf{S})$ is the regularization derived from prior information of $\mathbf{E}$ and $\mathbf{S}$. λ is a tradeoff scalar used to control the contribution of data fitting and regularization terms.

It is worth emphasizing that, in practice, $\mathbf{R}$ is relatively easy to be modeled and estimated according to the spectral specifications of corresponding sensors. However, the spatial degeneration operator $\mathbf{P}$ is rather difficult to be characterized, since several uncertainties remain, such as the PSF, blurring kernel size, and down-sampling mechanism.

III. Proposed Method

In this section, we introduce the subspace representation-based HSI super-resolution framework accounting for spectral variability. After that, we expatiate the detailed model specification of BSRSV. At the end of this section, we develop an effective inference scheme and analyze its computational complexity.

A. Spectral Variability in HSI–MSI Fusion

The HSI–MSI fusion issue formulated in (3) is based on the assumption that the observed images can relate to an identical latent HR-HSI $\mathbf{F}$ directly. However, the use of different sensors tends to introduce considerable spectral variability from a variety of causes: extrinsic (acquisition time, size of an object), intrinsic (morphology or composition of the material), environmental (illumination, atmospheric distortion), and so on. These effects cannot be adequately modeled by (1).

Given the presence of spectral variability between $\mathbf{H}$ and $\mathbf{M}$, the latent HR-HSI $\mathbf{F}$ in (1) are supposed to be different. Therefore, the observation model accounting for spectral variability can be modeled as

$$\begin{aligned}\mathbf{M}&=\mathbf{R}\mathbf{F}_m+\mathbf{N}_m\\ \mathbf{H}&=\mathbf{F}_h\mathbf{P}+\mathbf{N}_h\end{aligned} \quad (4)$$

where $\mathbf{F}_h \in \mathbb{R}^{L\times MN}$ and $\mathbf{F}_m \in \mathbb{R}^{L\times MN}$ correspond to latent HR-HSIs of observed $\mathbf{H}$ and $\mathbf{M}$, respectively. Then, we decompose $\mathbf{F}_m$ and $\mathbf{F}_h$ as

$$\mathbf{F}_m=\mathbf{E}_m\mathbf{S}^{\top},\quad \mathbf{F}_h=\mathbf{E}_h\mathbf{S}^{\top}. \quad (5)$$

In this scenario, the spectral dictionaries $\mathbf{E}_m$ and $\mathbf{E}_h$ can be different due to the existence of spectral variability. To simplify the representation model, we assumed that $\mathbf{F}_m$ and $\mathbf{F}_h$ share the identical texture feature $\mathbf{S}$.

Latest research studies have revealed that parametric models are flexible and physically reasonable to depict the varying spectra. Different parametric models have been recently proposed to account for spectral variability in hyperspectral unmixing [38]. Among them, the generalized linear mixture model (GLMM) [39] represents more complex spectral variability effects by considering an individual scaling factor for each spectral band of the endmembers. This model introduces a connection between the amount of spectral variability and the amplitude of reference spectral signatures, representing spectral mismatches caused by seasonal changes, global illumination, or atmospheric variations.

Motivated by the GLMM, we propose to characterize spectral variability by employing a set of nonnegative scaling factors for each spectral basis. In this manner, the spectral dictionary $\mathbf{E}_h$ is represented as a function of $\mathbf{E}_m$

$$\mathbf{E}_h=\mathbf{E}_m\odot\mathbf{T} \quad (6)$$

where $\mathbf{T}\in\mathbb{R}^{L\times K}$ denotes a scaling factor matrix and $\odot$ is the Hadamard product, namely, the entrywise product of two vectors, matrices, or tensors of the same size.

According to the above representation models, the HSI super-resolution in the presence of spectral mismatches is transformed as the estimation of $\mathbf{E}_m$, $\mathbf{T}$, and $\mathbf{S}$. However, as previously mentioned, the absence of $\mathbf{P}$ makes the estimation hard to be achieved.

It is common knowledge that the up-sampled LR-HSI (same spatial size as that of HR-HSI) can be regarded as the low-pass structures of HR-HSI [36], [40]. Consequently, we represent the up-sampled HSIs as

$$\widetilde{\mathbf{H}}=\mathbf{E}_h\mathbf{S}^{\top}\mathbf{P}\widetilde{\mathbf{P}}+\mathbf{N}_{\widetilde{h}} \quad (7)$$

where $\widetilde{\mathbf{H}}\in\mathbb{R}^{L\times MN}$ is the resampled HSI, $\widetilde{\mathbf{P}}\in\mathbb{R}^{mn\times MN}$ is an up-sampling operator, and $\mathbf{N}_{\widetilde{h}}$ is the additive noise and modeling error. For simplicity of notations, we denote $\mathbf{S}^{\top}\mathbf{P}\widetilde{\mathbf{P}}=\mathbf{A}^{\top}$, which represents the texture feature of low-frequency part. Accordingly, the coefficient matrix $\mathbf{S}^{\top}$ can be represented as the combination of two texture feature matrices $\mathbf{S}^{\top}=\mathbf{A}^{\top}+\mathbf{D}^{\top}$, namely $\mathbf{A}^{\top}$ and corresponding compensation components $\mathbf{D}^{\top}$.

Then, the observation models accounting for spectral mismatches in this work can finally be expressed as

$$\begin{aligned}\mathbf{M}&=\mathbf{R}\mathbf{E}\left(\mathbf{A}+\mathbf{D}\right)^{\top}+\mathbf{N}_m\\ \widetilde{\mathbf{H}}&=\left(\mathbf{E}\odot\mathbf{T}\right)\mathbf{A}^{\top}+\mathbf{N}_{\widetilde{h}}\end{aligned} \quad (8)$$

where $\mathbf{E}$ is the reference spectral dictionary.

B. Model Specification

As the observed collections $\widetilde{\mathbf{H}}$ and $\mathbf{M}$ are measured by disparate sensors at different time instants, the noise terms $\mathbf{N}_{\widetilde{h}}$ and $\mathbf{N}_m$ are assumed to be statistically independent. As a consequence, based on (8) and D-separation theory [41], we obtain the following likelihood function:

$$\begin{aligned}&p\left(\mathbf{M},\widetilde{\mathbf{H}}\middle|\mathbf{E},\mathbf{T},\mathbf{A}^{\top},\mathbf{D}^{\top}\right)\\ &\quad=p\left(\mathbf{M}\middle|\mathbf{E},\mathbf{A}^{\top},\mathbf{D}^{\top}\right)p\left(\widetilde{\mathbf{H}}\middle|\mathbf{E},\mathbf{T},\mathbf{A}^{\top}\right).\end{aligned} \quad (9)$$

In this research, we propose to obtain HR-HSI by estimating the spectral dictionary $\mathbf{E}$, scaling factors $\mathbf{T}$, and textures features $\mathbf{A}^{\top}$ and $\mathbf{D}^{\top}$. To this end, we give model specifications as follows.

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1) Spectral Dictionary: The spectral dictionary $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_K]$ is composed of several basis vectors containing spectral signatures, where K is the number of basis vectors and $\mathbf{e}_k$ is the kth basis vector. It is common knowledge that reflectance spectra are a relatively smooth function of wavelength [3]. The Gaussian process [42], [37] incorporates the regularly occurring smoothness of spectra providing an emerging tool for natural spectra characterization. Therefore, we employ Gaussian processes with appropriate kernels rather than simple multivariate Gaussian distributions for spectral modeling.

Specifically, each basis vector $\mathbf{e}_k$ is assumed to be a sample drawn from the corresponding Gaussian process as follows:

$$\mathbf{e}_k \sim \mathcal{GP}\left(\mathbf{e}_k \middle| \mathbf{e}_{k_0},\ \Sigma_k\right) \tag{10}$$

where $\mathbf{e}_{k_0} \in \mathbb{R}^L$ and $\Sigma_k \in \mathbb{R}^{L\times L}$ are the mean and kernel of $\mathbf{e}_k$, respectively, which are transformed to match the spectral quantization of HR-HSI.

To facilitate the relative smoothness of the spectra in the fused results, we promote the spectral correlation between consecutive bands of the HSI by defining the kernel as

$$\Sigma_k = \frac{1}{\eta_k}\exp\left(\frac{-|\theta_a - \theta_b|}{l^2}\right) \tag{11}$$

where l is the number of spectral bands of MSIs, and θ_a and θ_b correspond to the wavelengths at the ath and bth band of HSI. The scaling parameter η_k can be regarded as the precision of kernel Σ_k, which controls the average distance of the underlying function from the mean. The parameter η_k is determined automatically by placing a noninformative Gamma prior such that

$$\eta_k \sim \mathrm{Gam}(\eta_k \mid a_0,\ b_0) \tag{12}$$

where hyperparameters a_0 and b_0 are usually set to small values for introducing noninformative priors.

2) Scaling Factors: The scaling factor matrix $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_K]$ determines the amount of spectral variability in the reference spectral dictionary $\mathbf{E}$. In this research, $\mathbf{T}$ is constrained to be close to unity by employing the following distribution:

$$\mathbf{t}_k \sim \mathcal{N}\left(\mathbf{t}_k \middle| \mathbf{1}_L,\ \lambda_T^{-1}\mathbf{I}_L\right) \tag{13}$$

where $\mathbf{1}_L \in \mathbb{R}^L$ is a vector of all 1's, $\mathbf{I}_L$ is the $L \times L$ identity matrix, and λ_T is the precision parameter.

3) Texture Features: The texture feature matrices $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_K]$ and $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_K]$ contain the spatial structures of low- and high-frequency components in HR-HSI. As displayed in Fig. 1, $\mathbf{A}$ and $\mathbf{D}$ tend to exhibit distinct characteristics, and therefore, we model them separately.

The distribution of low-frequency texture message $\mathbf{A}$ can be relatively easy to be characterized, and we employ multivariate Gaussian distribution to model each vector of $\mathbf{A}$ as follows:

$$\mathbf{a}_k \sim \mathcal{N}\left(\mathbf{a}_k \middle| \mathbf{a}_{k_0}, \alpha_k^{-1}\mathbf{I}_{MN}\right) \tag{14}$$

where $\mathbf{a}_{k_0} \in \mathbb{R}^{MN}$ and α_k correspond to the mean vector and precision parameter of $\mathbf{a}_k$, respectively. We further place a conjugate Gamma prior over α_k with hyperparameters c_0 and d_0 as

$$\alpha_k \sim \mathrm{Gam}(\alpha_k | c_0,\ d_0). \tag{15}$$

According to Fig. 1, the texture feature of high-frequency components $\mathbf{D}$ tends to comply with the complex distribution. A simple Gaussian distribution is significantly limited when it comes to modeling those structures, whereas the combination of several Gaussian components can give a better characterization.

To impose a suitable prior on $\mathbf{D}$, we employ a mixture of Gaussians and assume that all the entries of the same column vector are drawn from R independent Gaussian distributions as follows:

$$\mathbf{d}_k \sim \sum_{r=1}^{R} \pi_r \mathcal{N}\left(\mathbf{d}_k \middle| \mathbf{d}_{kr0},\quad \beta_r^{-1}\mathbf{I}_{MN}\right) \tag{16}$$

where the mean and precision parameter of the rth component are denoted as $\mathbf{d}_{kr0}$ and β_r, respectively. The parameter $\pi_r \geqslant 0$ is a mixing coefficient that satisfies the requirement $\sum_{r=1}^{R} \pi_r = 1$. It has been proven that the linear superposition of sufficient Gaussian components with suitable means, covariances, and mixing coefficients can approximate almost any continuous density to arbitrary accuracy [37].

After introducing an R-dimensional vector $\mathbf{z}_k$, (16) can be reformulated as the following model [37]:

$$\mathbf{d}_k \sim \prod_{r=1}^{R} \mathcal{N}\left(\mathbf{d}_k \middle| \mathbf{d}_{kr0},\ \beta_r^{-1}\mathbf{I}_{MN}\right)^{z_{kr}} \tag{17}$$

where $\mathbf{z}_k \in \mathbb{R}^R$ is a binary random vector in which a specific entry is 1 and other entries are 0, namely, $\mathbf{z}_k = (z_{k1}, z_{k2}, \ldots, z_{kR}) \in \{0, 1\}^R$ and $\sum_{r=1}^{R} z_{kr} = 1$.

For the vector $\mathbf{z}_k$, we impose a multinomial prior over it as

$$\mathbf{z}_k \sim \mathrm{Mult}\left(\mathbf{z}_k | \boldsymbol{\pi}\right). \tag{18}$$

We further place conjugate priors over β_r and vector $\boldsymbol{\pi}$ as

$$\begin{aligned} \beta_r &\sim \mathrm{Gam}(\beta_r \mid e_0,\ f_0) \\ \boldsymbol{\pi} &\sim \mathrm{Dir}\left(\boldsymbol{\pi} \mid \mathbf{g}_0\right) \end{aligned} \tag{19}$$

where $\boldsymbol{\pi}$ complies with Dirichlet distribution with hyperparameter $\mathbf{g}_0 = [g_{01}, g_{02}, \ldots, g_{0R}]$ and β_r follows Gamma distribution with hyperparameters e_0 and f_0.

4) Noise Corruption: In this research, each element of the noise term $\mathbf{N}_m$ is presumed to comply with independent identically distributed Gaussian distribution, leading to $\mathbf{n}_i^m \sim \mathcal{N}(\mathbf{n}_i^m \mid \mathbf{0}_l,\ \lambda_M^{-1}\mathbf{I}_l)$ with noise precision λ_M and $i = 1, 2, \ldots, MN$. The noise matrix $\mathbf{N}_m$ can be equivalently reformulated according to the following matrix Gaussian distribution:

$$\mathbf{N}_m \sim \mathcal{MN}\left(\mathbf{N}_m \middle| \mathbf{0}_{l\times MN},\ \lambda_M^{-1}\mathbf{I}_l,\ \mathbf{I}_{MN}\right) \tag{20}$$

where $\mathbf{0}_{l\times MN} \in \mathbb{R}^{l\times MN}$ is the matrix composed of 0. Analogously, the noise matrix $\mathbf{N}_{\tilde{h}}$ can be represented as

$$\mathbf{N}_{\tilde{h}} \sim \mathcal{MN}\left(\mathbf{N}_{\tilde{h}} \middle| \mathbf{0}_{L\times MN},\ \lambda_H^{-1}\mathbf{I}_L,\ \mathbf{I}_{MN}\right). \tag{21}$$

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To make the assumption model more reliable, we also place hyperpriors over the noise precisions λ_M and λ_H, leading to

$$\begin{aligned}\lambda_M &\sim \mathrm{Gam}(\lambda_M \mid m_0,\, n_0)\\ \lambda_H &\sim \mathrm{Gam}(\lambda_H \mid p_0,\, q_0)\end{aligned} \tag{22}$$

where m_0, n_0, p_0, and q_0 are hyperparameters used to introduce noninformative priors.

According to the above likelihood and priors, we show the probabilistic graphical structure of the proposed model in Fig. 2. To simplify notations, the collection of unknowns is denoted together by $\boldsymbol{\Theta} = \{\mathbf{E}, \mathbf{T}, \mathbf{A}, \mathbf{D}, \boldsymbol{\eta}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\pi}, \{\mathbf{z}_k\}_{k=1}^K, \lambda_H, \lambda_M\}$. The joint distribution of this model can be expressed by

$$\begin{aligned}p\big(\mathbf{M}, \widetilde{\mathbf{H}}, \boldsymbol{\Theta}\big) = {} & p\big(\mathbf{M}, \widetilde{\mathbf{H}} | \boldsymbol{\Theta}\big) p\big(\mathbf{E}|\boldsymbol{\eta}\big) p\big(\mathbf{T}\big) p\big(\mathbf{A}|\boldsymbol{\alpha}\big)\\ & \times p\Big(\mathbf{D}\Big|\boldsymbol{\beta}, \big\{\mathbf{z}_k\big\}_{k=1}^K\Big) \prod_{k=1}^K p\big(\mathbf{z}_k|\boldsymbol{\pi}\big)\\ & \times p(\boldsymbol{\eta}) p(\boldsymbol{\alpha}) p(\boldsymbol{\beta}) p(\boldsymbol{\pi}) p(\lambda_M) p(\lambda_H).\end{aligned} \tag{23}$$

C. Bayesian Inference

The proposed BSRSV is fully conjugate, which means we can derive analytical expressions for posterior distributions. We employ Gibbs sampling technique [37] to converge the posterior distribution of the individual factor by drawing samples from corresponding probability distribution in turn. In the following equations, the shorthand notation of expectation with respect to a latent variable is denoted as $\langle\cdot\rangle$, and the ℓ_2-norm of the vector is represented as $\|\cdot\|_2$. The contribution of the kth spectral basis $\mathbf{e}_k$ to $\mathbf{M}$ and $\widetilde{\mathbf{H}}$ are denoted as

$$\begin{aligned}\mathbf{M}_{e_k} &= \mathbf{M} - \mathbf{R}\mathbf{E}\big(\mathbf{A}+\mathbf{D}\big)^\top + \mathbf{R}\mathbf{e}_k\big(\mathbf{a}_k + \mathbf{d}_k\big)^\top\\ \widetilde{\mathbf{H}}_{e_k} &= \widetilde{\mathbf{H}} - \big(\mathbf{E}\odot\mathbf{T}\big)\mathbf{A}^\top + \big(\mathbf{e}_k \odot \mathbf{t}_k\big)\mathbf{a}_k^\top.\end{aligned} \tag{24}$$

According to the above notations, we sequentially draw samples from each variable in $\boldsymbol{\Theta}$ and keep iterating until convergence. The analytical expressions derived by Gibbs samplers are given as follows.

1) *Sample for* $\mathbf{e}_k$*:* Exploiting the linear Gaussian model [43], it can be shown that $\mathbf{e}_k$ is drawn from $\mathcal{GP}(\mathbf{e}_k \mid \langle\mathbf{e}_k\rangle,\, \widehat{\Sigma}_k)$ where

$$\begin{aligned}\widehat{\Sigma}_k = {} & \Big(\lambda_H\big\|\mathbf{a}_k\big\|_2^2 \mathrm{diag}\big(\mathbf{t}_k\big)\mathrm{diag}\big(\mathbf{t}_k\big)\\ & + \lambda_M\big\|\mathbf{a}_k + \mathbf{d}_k\big\|_2^2 \mathbf{R}^\top\mathbf{R} + \Sigma_k^{-1}\Big)^{-1}\end{aligned} \tag{25}$$

$$\begin{aligned}\big\langle\mathbf{e}_k\big\rangle = {} & \widehat{\Sigma}_k\Big(\lambda_H \mathrm{diag}\big(\mathbf{t}_k\big)\widetilde{\mathbf{H}}_{e_k}\mathbf{a}_k\\ & + \lambda_M\mathbf{R}^\top\mathbf{M}_{e_k}\big(\mathbf{a}_k + \mathbf{d}_k\big) + \Sigma_k^{-1}\mathbf{e}_{k0}\Big).\end{aligned} \tag{26}$$

2) *Sample for* η_k*:* For brevity, we denote $\Sigma_k = \eta_k^{-1}\Sigma_E$. Thus, η_k can be drawn from a Gamma distribution $\mathrm{Gam}(\eta_k \mid a_k,\, b_k)$ with parameters

$$\begin{aligned}a_k &= a_0 + \frac{L}{2}\\ b_k &= b_0 + \frac{\big(\mathbf{e}_k^\top - \mathbf{e}_{k_0}^\top\big)\Sigma_E^{-1}\big(\mathbf{e}_k - \mathbf{e}_{k_0}\big)}{2}.\end{aligned} \tag{27}$$

3) *Sample for* $\mathbf{t}_k$*:* $\mathbf{t}_k$ complies with the multivariate Gaussian distribution $\mathcal{N}(\mathbf{t}_k \mid \langle\mathbf{t}_k\rangle,\, \Lambda_k)$ with parameters

$$\begin{aligned}\Lambda_k &= \Big(\lambda_H\big\|\mathbf{a}_k\big\|_2^2 \mathrm{diag}\big(\mathbf{e}_k\big)\mathrm{diag}\big(\mathbf{e}_k\big) + \lambda_T\mathbf{I}_L\Big)^{-1}\\ \big\langle\mathbf{t}_k\big\rangle &= \Lambda_k\Big(\lambda_H \mathrm{diag}\big(\mathbf{e}_k\big)\widetilde{\mathbf{H}}_{e_k}\mathbf{a}_k + \lambda_T\mathbf{1}_L\Big).\end{aligned} \tag{28}$$

4) *Sample for* $\mathbf{a}_k$*:* by employing the conjugacy of the probability distribution, we can sample $\mathbf{a}_k$ form $\mathcal{N}(\mathbf{a}_k \mid \langle\mathbf{a}_k\rangle,\, \Phi_k)$, where

$$\Phi_k = \Big(\lambda_H\big\|\mathbf{e}_k \odot \mathbf{t}_k\big\|_2^2 + \lambda_M\big\|\mathbf{R}\mathbf{e}_k\big\|_2^2 + \alpha_k\Big)^{-1}\mathbf{I}_{MN} \tag{29}$$

$$\begin{aligned}\big\langle\mathbf{a}_k\big\rangle = {} & \Phi_k\Big(\lambda_H\widetilde{\mathbf{H}}_{e_k}^\top\big(\mathbf{e}_k \odot \mathbf{t}_k\big)\\ & + \lambda_M\big(\mathbf{M}_{e_k} - \mathbf{R}\mathbf{e}_k\mathbf{d}_k^\top\big)^\top\mathbf{R}\mathbf{e}_k + \alpha_k\mathbf{a}_{k0}\Big).\end{aligned} \tag{30}$$

5) *Sample for* α_k*:* α_k can be sampled from a Gamma distribution $\mathrm{Gam}(\alpha_k \mid c_k,\, d_k)$ with parameters

$$\begin{aligned}c_k &= c_0 + \frac{MN}{2}\\ d_k &= d_0 + \frac{\big\|\mathbf{a}_k - \mathbf{a}_{k_0}\big\|_2^2}{2}.\end{aligned} \tag{31}$$

6) *Sample for* $\mathbf{d}_k$*:* By employing the linear Gaussian model, we can find that $\mathbf{d}_k$ tend to comply with multivariate Gaussian distribution $\mathcal{N}(\mathbf{d}_k \mid \langle\mathbf{d}_k\rangle,\, \Psi_k)$, where

$$\begin{aligned}\Psi_k &= \left(\lambda_M\big\|\mathbf{R}\mathbf{e}_k\big\|_2^2 + \sum_{r=1}^R z_{kr}\beta_r\right)^{-1}\mathbf{I}_{MN}\\ \big\langle\mathbf{d}_k\big\rangle &= \Psi_k\Big(\lambda_M\big(\mathbf{M}_{e_k} - \mathbf{R}\mathbf{e}_k\mathbf{a}_k^\top\big)^\top\mathbf{R}\mathbf{e}_k\Big).\end{aligned} \tag{32}$$

7) *Sample for* β_r*:* Guided by the conjugacy between the Gaussian and the Gamma distributions, we can sample β_r from Gamma distribution $\mathrm{Gam}(\beta_r \mid e_r,\, f_r)$, where

$$\begin{aligned}e_r &= e_0 + \frac{\sum_{k=1}^K z_{kr}MN}{2}\\ f_r &= f_0 + \frac{\sum_{k=1}^K z_{kr}\big\|\mathbf{d}_k\big\|_2^2}{2}.\end{aligned} \tag{33}$$

8) *Sample for* $\mathbf{z}_k$*:* $\mathbf{z}_k$ can be drawn from the following multinomial distribution:

$$\begin{aligned}&\mathbf{z}_k \sim \mathrm{Mult}\big(\mathbf{z}_k|\widetilde{\boldsymbol{\pi}}\big)\\ &\widetilde{\pi}_r = \pi_r\prod_{i=1}^{MN}\mathcal{N}\big(d_{ik}|0,\beta_r^{-1}\big)\sum_{t=1}^R\pi_t\prod_{i=1}^{MN}\mathcal{N}\big(d_{ik}|0,\beta_t^{-1}\big)\end{aligned} \tag{34}$$

where $\widetilde{\boldsymbol{\pi}} = (\widetilde{\pi}_1, \widetilde{\pi}_2, \ldots, \widetilde{\pi}_R)$ and $\widetilde{\pi}_r$ is the rth entry of $\widetilde{\boldsymbol{\pi}}$.

9) *Sample for* $\boldsymbol{\pi}$*:* $\boldsymbol{\pi}$ can be sampled from the following Dirichlet distribution:

$$\boldsymbol{\pi} \sim \mathrm{Dir}\big(\boldsymbol{\pi} \mid \mathbf{g}\big) \tag{35}$$

where $\mathbf{g} = (g_1, g_2, \ldots, g_R)$ with entry $g_r = \sum_{k=1}^K z_{kr} + g_{0r}$.

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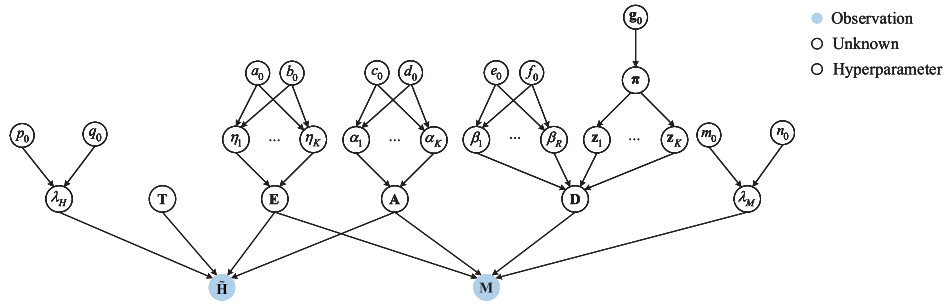


Fig. 2. Probabilistic graphical model of the proposed BSRSV.

10) *Sample for* λ_M*:* We can show that the posterior is a gamma distribution, that is, $p(\lambda_M) = \text{Gam}(\lambda_M \mid m, n)$, where the posterior parameters can be updated by

$$m = m_0 + \frac{MNl}{2}$$
$$n = n_0 + \frac{\left\|\mathbf{M} - \mathbf{RE}(\mathbf{A}+\mathbf{D})^\top\right\|_F^2}{2}. \tag{36}$$

11) *Sample for* λ_H*:* Similarly, the posterior is a gamma distribution, that is, $p(\lambda_H) = \text{Gam}(\lambda_H \mid p, q)$, where the posterior parameters can be updated by

$$p = p_0 + \frac{MNL}{2}$$
$$q = q_0 + \frac{\left\|\widetilde{\mathbf{H}} - (\mathbf{E}\odot\mathbf{T})\mathbf{A}^\top\right\|_F^2}{2}. \tag{37}$$

The complete inference algorithm of BSRSV is summarized in Algorithm 1.

Algorithm 1 BSRSVfor HSI Super-Resolution

Input: $\widetilde{\mathbf{H}}$, $\mathbf{M}$, $\mathbf{R}$, R, K

Initialization: $\left\{\mathbf{e}_{k0}, \mathbf{a}_{k0}, \mathbf{d}_{k0}\right\}_{k=1}^{K}$, a_0, b_0, c_0, d_0, e_0, f_0, $\mathbf{g}_0$, p_0, q_0, m_0, n_0, λ_T.
while not converged and $iter < maxIter$ **do**
  for $k = 1 : K$ **do**
    Update the posterior of $\mathbf{e}_k$ by (25) and (26);
    Update the posterior of η_k by (27);
    Update the posterior of $\mathbf{t}_k$ by (28);
    Update the posterior of $\mathbf{a}_k$ by (29) and (30);
    Update the posterior of α_k by (31);
    Update the posterior of $\mathbf{d}_k$ by (32);
    Update the posterior of $\mathbf{z}_k$ by (34);
  end for
  Update the posterior of $\boldsymbol{\beta}$ by (33);
  Update the posterior of $\boldsymbol{\pi}$ by (35);
  Update the posterior of λ_M by (36);
  Update the posterior of λ_H by (37);
  Check the convergence conditions;
  $iter \leftarrow iter + 1$;
end while
Reconstruct $\mathbf{F}_m$ and $\mathbf{F}_h$ by using (5).

Output: $\mathbf{F}_m$ or $\mathbf{F}_h$

D. Initialization

The Gibbs sampling can approximate the latent posterior probability distributions effectively. However, in the proposed BSRSV, several hyperparameters are required to start the sampling procedure. Following [44], [45], [46], the top-level hyperparameters including a_0, b_0, c_0, d_0, e_0, f_0, p_0, q_0, m_0, and n_0 are set to 10^{-6} for introducing noninformative priors to the precision parameters. To facilitate the nonnegativity of $\{\mathbf{a}_{k0}\}_{k=1}^K$ and relative smoothness of spectra $\{\mathbf{e}_{k0}\}_{k=1}^K$, we restrict them to be estimated from $\widetilde{\mathbf{H}}$ using online dictionary learning algorithm [47].

It is noteworthy that the high-frequency structure of HR-HSI is not nonnegative, which means nonnegative prior cannot be placed over $\{\mathbf{d}_{k0}\}_{k=1}^K$. For tractability, zero vectors are used for $\{\mathbf{d}_{k0}\}_{k=1}^K$. The number of mixture components R is fixed as 3, which can also be selected automatically by the parameter selection mechanism proposed in [46]. The precision parameter λ_T is set as L to characterize the variations in spectral signatures. However, when the influence of spectral variability is negligible, λ_T is set as infinity to disable (13).

E. Computational Complexity

The approximations of the above posterior distributions have different orders of computational complexity. Among them, the spectral dictionary $\mathbf{E}$, scaling factors $\mathbf{T}$, and texture feature matrices $\mathbf{A}$ and $\mathbf{D}$ take most of the computational resources. Specifically, $\mathbf{E}$ and $\mathbf{T}$ have the same computational cost $\mathcal{O}(KL^3 + KMNL)$. While the computational cost of $\mathbf{A}$ and $\mathbf{D}$ is $\mathcal{O}(KLl + KMNl)$. For the update of remaining unknowns, the computational costs are $\mathcal{O}(KL^2)$ for $\boldsymbol{\eta}$, $\mathcal{O}(KMN)$ for $\boldsymbol{\alpha}$, $\mathcal{O}(MNR)$ for $\mathbf{z}$, $\mathcal{O}(KR)$ for $\boldsymbol{\pi}$, $\mathcal{O}(KMNl)$ for λ_M, and $\mathcal{O}(KMNL)$ for λ_H. Since $l \ll L$, $R \ll L$, the overall complexity of our algorithm is $\mathcal{O}(K(L^3 + MNL))$, which scales linearly with the number of spectral signatures but polynomially with the number of spectral bands.

IV. Experimental Study

In this section, we devise four experiments to give a comprehensive demonstration of BSRSV in terms of fusion performance and robustness to spectral variability. The first experiment evaluates the robustness of spectral variability by using synthesized images with controlled spectral mismatches. The second one employs semireal images with no spectral

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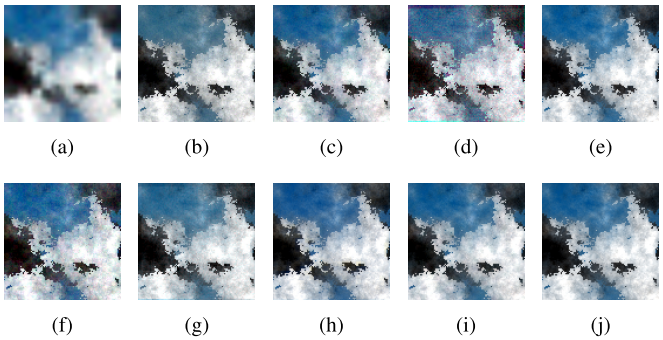


Fig. 3. Visual inspection for the reconstructed $\mathbf{F}_m$ of the synthetic dataset with controlled spectral mismatches. (a) LR-HSI. (b) BSR. (c) CMS. (d) LTMR. (e) MDF. (f) IR-TenSR. (g) CNNFUS. (h) FuVar. (i) BSRSV. (j) GT.

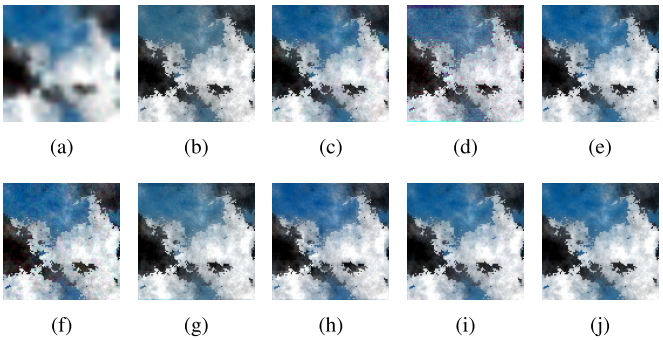


Fig. 4. Visual inspection for the reconstructed $\mathbf{F}_h$ of the synthetic dataset with controlled spectral mismatches. (a) LR-HSI. (b) BSR. (c) CMS. (d) LTMR. (e) MDF. (f) IR-TenSR. (g) CNNFUS. (h) FuVar. (i) BSRSV. (j) GT.

variability using different spatial degeneration methods. The third example fuses the real-life HSIs and MSIs observed at distinct time instants, which means the influence of spectral variability should be taken into consideration. The last one also employs real HSIs and MSIs, while these images were observed at the same condition and time instant, thus no spectral variability is contained in these images.

A. Datasets

In this research, synthetic and four real-life datasets are employed as benchmarks for experiments. In what follows, we give the details of these datasets.

1) Synthetic Dataset With Controlled Spectral Mismatches: This dataset was initially released in FuVar [35] and contained a pair of HR-HSIs $\mathbf{F}_m$ and $\mathbf{F}_h$ with identical size $100 \times 100 \times 224$. In this dataset, $\mathbf{S}$ is the coefficient matrix of $\mathbf{F}_m$ and $\mathbf{F}_h$, which was generated with a Gaussian random field. The reference spectral dictionary $\mathbf{E}$ is composed of three spectral samples extracted from the United States Geological Survey (USGS) spectra library. The scaling factor matrix $\mathbf{T}$ was generated with random piecewise linear functions centered around a unitary gain. The HR-HSIs $\mathbf{F}_m$ and $\mathbf{F}_h$ were constructed with these data according to (5) and (6).

2) University of Pavia Dataset: This image covers the University of Pavia, which was obtained by the Reflective Optics System Imaging Spectrometer [48]. The spatial size of this dataset is 610×340 with 115 spectral bands covering 0.43–0.86 μ m. In our experiment, we excluded the bands contaminated by water vapor absorption and kept the top $320 \times 320 \times 93$ section as ground truth.

3) Lockwood Dataset: This dataset was given in FuVar [35] and contained a pair of HSIs and MSIs captured over the Lockwood area. The HSIs and MSIs were obtained by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) on 2018-08-20 and Sentinel-2A instrument on 2018-10-19, respectively. After excluding the noisy bands, the size of the HSI is reduced to $80 \times 80 \times 173$, and the MSI is of size $80 \times 80 \times 10$.

4) Ivanpah Playa Dataset: This dataset was also presented in [35] and composed of a pair of HSIs and MSIs captured over the Ivanpah Playa area. The HSI, with 80×80 pixels and 173 spectral bands, was acquired by AVIRIS on 2015-10-24. The corresponding MSI was obtained by Sentinel-2A instrument with size of $80 \times 80 \times 10$ on 2017-12-17.

TABLE I

QUANTITATIVE EVALUATION FOR THE RECONSTRUCTED $\mathbf{F}_m$ OF THE SYNTHETIC DATASET WITH CONTROLLED SPECTRAL MISMATCHES

Method	PSNR	ERGAS	SAM	UIQI
Reference	∞	0	0	1
BSR	32.2139	1.2160	2.4726	0.9491
CMS	35.4757	0.6417	0.8073	0.9618
LTMR	29.1079	1.3126	1.8271	0.9155
MDF	31.3630	1.1203	1.9455	0.9441
IR-TenSR	27.2565	1.8479	2.1239	0.9173
CNNFUS	29.9124	1.2368	1.5433	0.9516
FuVar	38.0362	0.4380	0.3506	**0.9850**
BSRSV	**38.1696**	**0.4333**	**0.3277**	0.9845

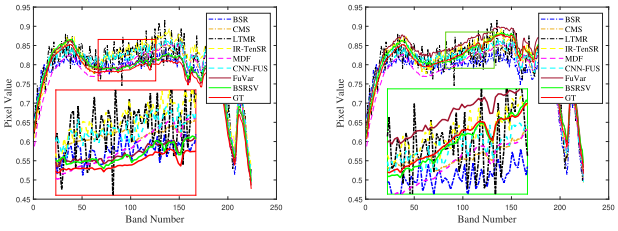


Fig. 5. Spectral curves of the selected pixel in the reconstructed HR-HSIs for the synthetic dataset. (a) Reconstruction for $\mathbf{F}_m$. (b) Reconstruction for $\mathbf{F}_h$.

5) NCALM Dataset: This dataset was obtained by the National Center for Airborne Laser Mapping (NCALM) and released for the 2018 IEEE GRSS Data Fusion Contest [49]. The NCALM dataset contains an HSI of size $1202 \times 4172 \times 50$ and an MSI of size $24\,040 \times 83\,440 \times 3$, which were taken by ITRES CASI 1500 and DiMAC ULTRALIGHT+ imaging systems, respectively. In our experiment, we chose an area of $30 \times 30 \times 48$ from HSIs and corresponding $600 \times 600 \times 3$ section from MSIs.

B. Comparison Methods and Evaluation Criterion

A set of baseline algorithms are employed for comparison, namely, FuVar [35], subspace-based low tensor multirank (LTMR) [50], iterative regularization method based on

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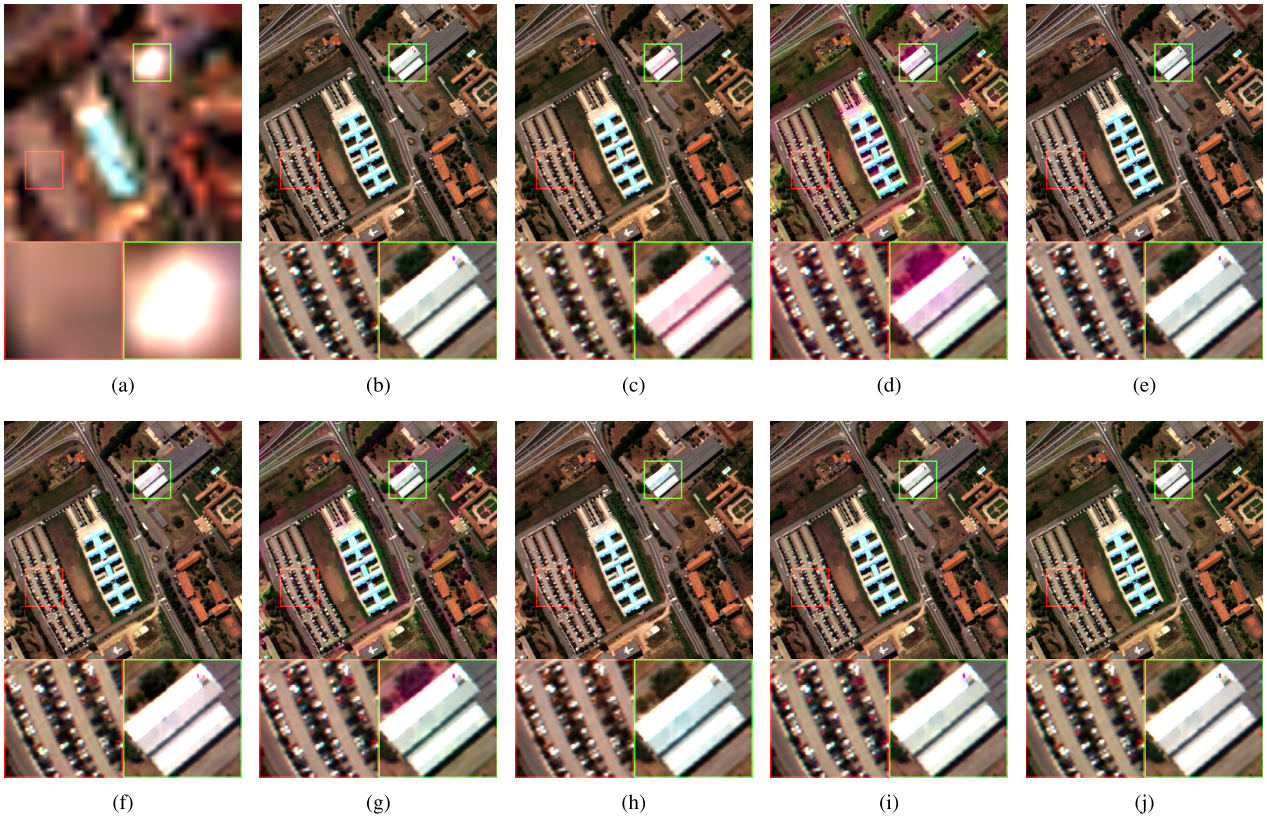


Fig. 6. Visual quality comparison for the reconstructed images of the Pavia dataset using averaging blurs in spatial degeneration. (a) LR-HSI. (b) BSR. (c) CMS. (d) LTMR. (e) MDF. (f) IR-TenSR. (g) CNNFUS. (h) FuVar. (i) BSRSV. (j) GT.

TABLE II

QUANTITATIVE EVALUATION FOR THE RECONSTRUCTED $\mathbf{F}_h$ OF THE SYNTHETIC DATASET WITH CONTROLLED SPECTRAL MISMATCHES

Method	PSNR	ERGAS	SAM	UIQI
Reference	∞	0	0	1
BSR	29.3999	1.5803	2.6461	0.9386
CMS	31.8875	1.2115	1.7493	0.9480
LTMR	33.0271	0.8179	1.4000	0.9282
MDF	31.8052	0.8747	**0.2509**	0.9858
IR-TenSR	35.5236	0.5489	0.8019	0.9621
CNNFUS	34.7045	0.6840	1.0258	0.9608
FuVar	36.3692	0.5653	0.4617	0.9748
BSRSV	**38.8471**	**0.3850**	0.3849	**0.9914**

tensor subspace representation (IR-TenSR) [51], BSR [25], matrix decomposition fusion (MDF) [52], clustering manifold structure-based method (CMS) [53], and CNNFUS [33]. Among them, the FuVar and MDF are guided by matrix factorization; IR-TenSR and LTMR are based on tensor decomposition; BSR is based on Bayesian theory; CNNFUS is a deep-learning-based method and CMS is a manifold-based approach. In this research, all the comparing methods are implemented with their released codes. We further reset several parameters for best performances.

For numeric assessment, we employ four conventional evaluation criteria, including the Erreur Relative Globale

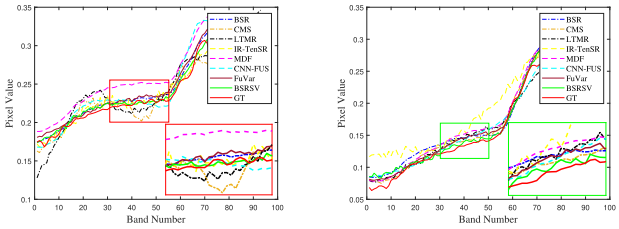


Fig. 7. Spectral curves of the selected pixel in the reconstructed HR-HSIs for the Pavia dataset. (a) Spatial degeneration with averaging blurs. (b) Spatial degeneration with Gaussian blurs.

Adimensionnelle de Synthèse (ERGAS), spectral angle mapper (SAM), peak signal-to-noise ratio (PSNR), and universal image quality index (UIQI) to evaluate the quality of reconstructions. PSNR can reveal the numerical similarity between ground truth and fusion outcome. SAM evaluates the degree of spectral angle between the result and reference image. ERGAS measures the universal quality of fusion result and the UIQI evaluates fusion performance by combining correlation error and contrast distortion.

C. Results on Synthesized Dataset With Spectral Variability

This experiment was performed to compare the above methods under a controlled spectral variability environment. In this section, we produce synthetic images according to Wald's protocol [54]. The HR-MSI $\mathbf{M}$ was simulated by down-sampling the spectral bands of $\mathbf{F}_m$ according to the

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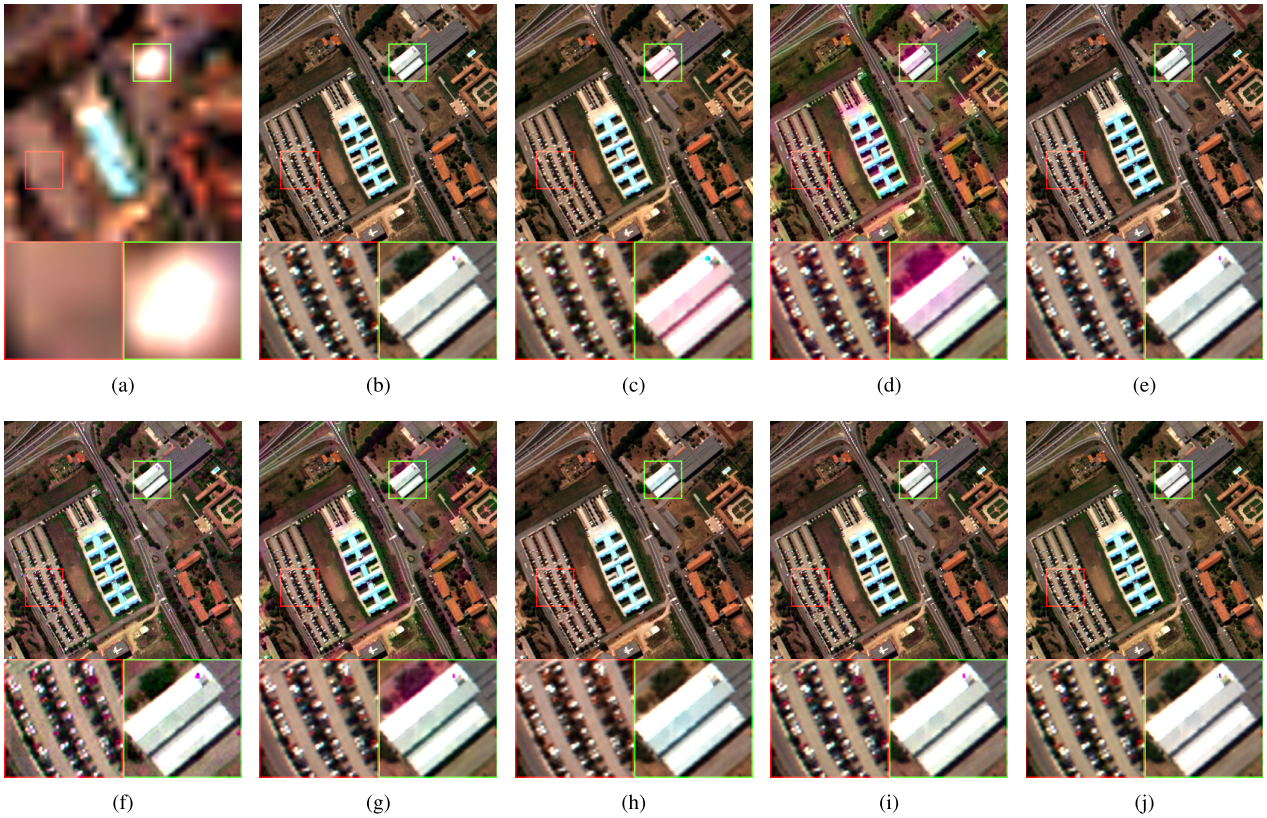


Fig. 8. Visual quality comparison for the reconstructed images of the Pavia dataset using Gaussian blurs in spatial degeneration. (a) LR-HSI. (b) BSR. (c) CMS. (d) LTMR. (e) MDF. (f) IR-TenSR. (g) CNNFUS. (h) FuVar. (i) BSRSV. (j) GT.

TABLE III

QUANTITATIVE EVALUATION FOR THE RECONSTRUCTED RESULTS OF THE SYNTHETIC IMAGES USING THE PAVIA DATASET AND AVERAGE BLUR

Method	PSNR	ERGAS	SAM	UIQI
Reference	∞	0	0	1
BSR	39.2368	0.4249	2.7523	0.9893
CMS	35.1713	0.7079	4.6968	0.9503
LTMR	33.1751	0.9457	6.7258	0.9259
MDF	33.4156	0.7303	2.6359	0.9809
IR-TenSR	32.8521	1.0874	5.9278	0.9162
CNNFUS	36.8877	0.5912	3.9711	0.9786
FuVar	37.1785	0.5345	3.5159	0.9814
BSRSV	**39.6882**	**0.3944**	**2.6278**	**0.9898**

TABLE IV

QUANTITATIVE EVALUATION FOR THE RECONSTRUCTED RESULTS OF THE SYNTHETIC IMAGES USING THE PAVIA DATASET AND GAUSSIAN BLUR

Method	PSNR	ERGAS	SAM	UIQI
Reference	∞	0	0	1
BSR	38.5038	0.4693	3.0621	0.9877
CMS	35.4967	0.7264	5.1321	0.9521
LTMR	33.6997	0.8707	6.4449	0.9344
MDF	37.9695	0.4607	**2.4863**	0.9890
IR-TenSR	32.4905	0.8851	5.7452	0.9450
CNNFUS	37.6102	0.5331	3.6911	0.9821
FuVar	33.3204	0.8626	5.2348	0.9616
BSRSV	**39.4589**	**0.4085**	2.7139	**0.9897**

spectral specifications of imaging sensors. The LR-HSI $\mathbf{H}$ was then generated by averaging and down-sampling over adjacent $sf \times sf$ blocks of $\mathbf{F}_h$, where the scaling factor $sf = 4$. We further introduce Gaussian noise to the generated HR-MSI and LR-HSI, resulting in a signal-to-noise ratio (SNR) of 30 dB for both images.

This experiment aims at estimating $\mathbf{F}_m$ and $\mathbf{F}_h$ simultaneously from $\mathbf{H}$ and $\mathbf{M}$. Figs. 3 and 4 show the outcomes of the super-resolved HR-HSIs $\mathbf{F}_m$ and $\mathbf{F}_h$, respectively. The numerical evaluation of outcomes with respect to $\mathbf{F}_m$ and $\mathbf{F}_h$ are presented in Tables I and II. We can find that BSRSV outperforms other methods in most of the quantitative metrics. In terms of PSNR, the performance gain achieved by BSRSV is larger than 2.5 dB for $\mathbf{F}_h$. Visually, we can find that BSRSV tends to produce the most promising outcomes, which makes a reasonable balance between the smooth regions and sharp discontinuities with inconspicuous spectral distortions and blurring artifacts. To illustrate the performance of fusion results in spectral preservation, we display the spectral curve of pixel [35, 35] in Fig. 5. We can find that BSRSV produces the most accurate results in both cases.

D. Results on Semireal Dataset Without Spectral Variability

This section was given to evaluate the robustness of methods to different spatial degeneration when there are no variations in spectral signatures between HSIs and MSIs. The first spatial

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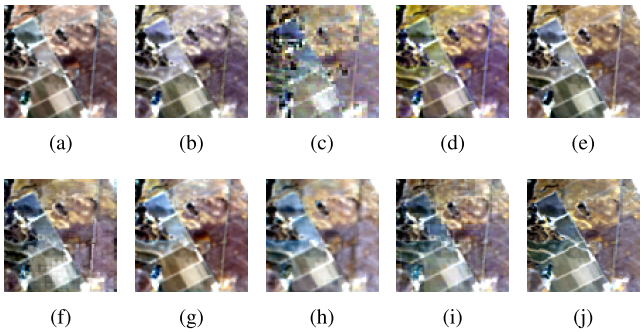


Fig. 9. False color images of the reconstructed results for the Lockwood dataset. (a) HR-MSI. (b) BSR. (c) CMS. (d) LTMR. (e) MDF. (f) IR-TenSR. (g) CNNFUS. (h) FuVar. (i) BSRSV. (j) GT.

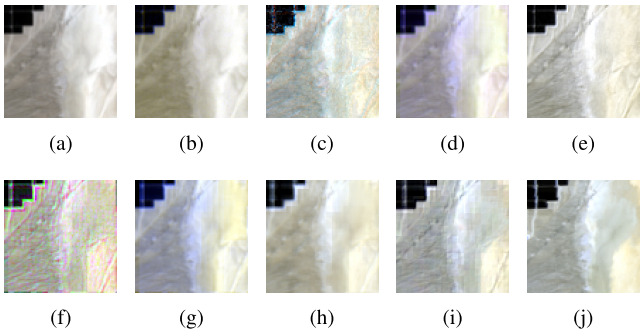


Fig. 10. False color images of the reconstructed results for the Playa dataset. (a) HR-MSI. (b) BSR. (c) CMS. (d) LTMR. (e) MDF. (f) IR-TenSR. (g) CNNFUS. (h) FuVar. (i) BSRSV. (j) GT.

TABLE V

PERFORMANCES OF THE HSI SUPER-RESOLUTION METHODS ON THE LOCKWOOD DATASET

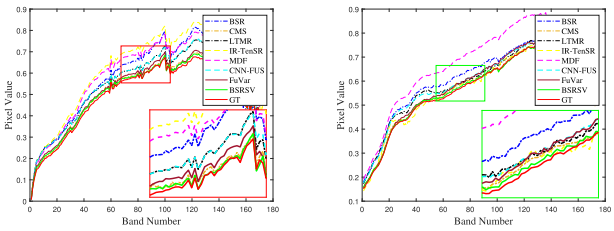
Method	PSNR	ERGAS	SAM	UIQI
Reference	∞	0	0	1
BSR	19.0431	6.2736	3.3100	0.6365
CMS	25.5557	4.0241	2.1815	0.5899
LTMR	25.2665	4.1244	3.9364	0.8137
MDF	20.1104	5.5768	1.9390	0.7510
IR-TenSR	24.0440	6.3528	5.1907	0.6016
CNNFUS	24.6764	4.2868	3.1564	0.7972
FuVar	29.3078	2.9066	2.1905	0.8059
BSRSV	**31.4052**	**2.4877**	**1.6232**	**0.8820**

degeneration method blurred the HR-HSI with an averaging operator and down-sampled it by a scaling factor of 16. The 4-band HR-MSI $\mathbf{M}$ was generated according to the spectral specifications of the IKONOS multispectral sensor. Gaussian noises were introduced into the generated $\mathbf{H}$ and $\mathbf{M}$, resulting in an SNR of 25 dB for both images. Fig. 6 shows the fusion results of above $\mathbf{H}$ and $\mathbf{M}$. Two regions of size 50×50 are enlarged and shown at the bottom of corresponding images. The quantitative evaluations are presented in Table III. We can find that BSRSV outperforms other methods in all metrics. The BSR also achieves competitive result with sharp edges and inconspicuous spectral distortions. We also show the spectral

TABLE VI

PERFORMANCES OF THE HSI SUPER-RESOLUTION METHODS ON THE PLAYA DATASET

Method	PSNR	ERGAS	SAM	UIQI
Reference	∞	0	0	1
BSR	20.3045	3.9910	2.3123	0.6418
CMS	27.5688	1.9896	1.9467	0.7646
LTMR	25.0873	2.7009	3.3617	0.8479
MDF	20.3002	3.8244	1.2278	0.7907
IR-TenSR	25.6706	2.4358	1.8472	0.7490
CNNFUS	25.2078	2.6001	2.9226	0.8533
FuVar	28.9131	1.6432	1.5079	0.8777
BSRSV	**32.2271**	**1.1177**	**0.9527**	**0.9347**



(a) (b)

Fig. 11. Spectral curves of the selected pixel in the reconstructed HR-HSI for the synthetic dataset. (a) Lockwood dataset. (b) Playa dataset.

curve of pixel [40, 40] in Fig. 7(a) for visual inspection. It can be found that BSRSV performs best in most spectral bands.

The second spatial degeneration scheme generated LR-HSI by blurring HR-HSI with a Gaussian filter and down-sampling it by a scaling factor of 16. The HR-MSI and Gaussian noises in this experiment were the same as that of the previous one. The fusion results are displayed in Fig. 8. As we can visually inspect from these results, IR-TenSR introduces blurring artifacts, and LTMR and FuVar exhibit different degrees of spectral distortions. We also present the numerical results in Table IV. In this experiment, BSRSV outperforms other methods in all quality evaluations except SAM. Fig. 7(b) gives the spectral curve of pixel [15, 15], which further validates the superiority of the proposed BSRSV in terms of spectral preservation. It is noteworthy that the spatial degeneration operator $\mathbf{P}$ in the competing methods is determined by the corresponding spatial degeneration scheme, whereas in the proposed BSRSV and BSR, the operator $\mathbf{P}$ is not required.

By comparing the visual and quantitative results in this section, we can conclude that the proposed BSRSV is robust to the spatial degeneration scheme and tend to give the best results in both cases.

E. Results on Real-Life Dataset With Spectral Variability

In this section, we measure the fusion results of different methods over two real-world datasets, namely, Lockwood and Playa datasets. Each dataset contains a pair of MSIs and HSIs obtained at different time instants, thus containing spectral mismatches. In this experiment, the observed HSIs

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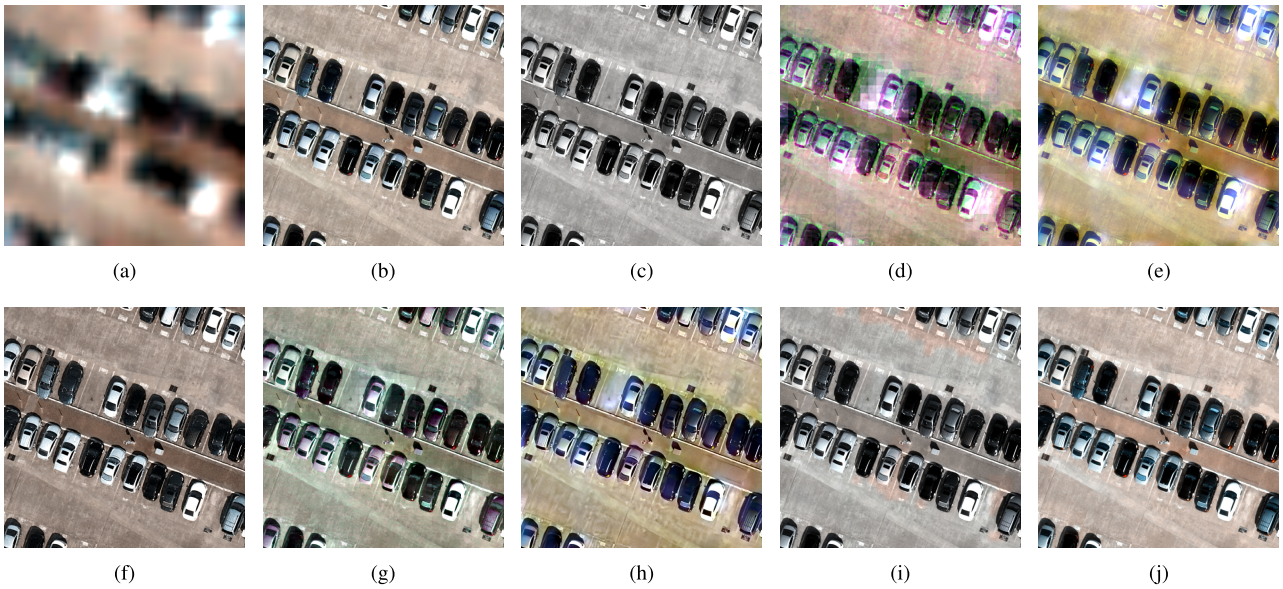


Fig. 12. RGB images of the real dataset and reconstructed results using the compared methods. (a) LR-HSI. (b) HR-MSI. (c) BSR. (d) CMS. (e) LTMR. (f) MDF. (g) IR-TenSR. (h) CNNFUS. (i) FuVar. (j) BSRSV.

TABLE VII
COMPUTATIONAL TIME OF THE COMPARED METHODS (SECONDS)

Method	BSR	CMS	LTMR	MDF	IR-TenSR	CNNFUS	FuVar	BSRSV
Pavia	536	136	204	27	278	75	359	327
NCALM	3761	221	762	59	648	182	2114	938

were blurred with an averaging operator and down-sampled by a scaling factor of 4. In this manner, the observed HSIs can be regarded as a reference. We further introduce Gaussian noise to the observed MSI and resampled HSI, resulting in the SNR of 30 dB for both images.

The quantitative evaluations presented in Tables V and VI clearly demonstrate the fact that the proposed BSRSV constantly achieves better results over other methods in all metrics. Visual inspection of the pseudo-RGB images given in Fig. 9 reveals a significant performance improvement achieved by BSRSV. Among other methods, the LTMR and MDF produce sharp edges at the cost of introducing spectral distortions. In the Playa dataset, both spectral and spatial mismatches are widely existed due to the large acquisition time difference between HSIs and MSIs. Therefore, the fusion results in Fig. 10 exhibit spatial structures that are not included in the reference image, such as right-bottom region. As we can inspect form Fig. 10, the most precise result is produced by the proposed BSRSV without introducing obvious spatial deterioration and spectral distortions. The spectral curves shown in Fig. 11 further support our visual inspection and demonstrate the superiority of BSRSV.

F. Results on Real-Life Dataset Without Spectral Variability

This example aims at validating the capacity of BSRSV in terms of practical applications when the influence of spectral variability is negligible. The NCALM dataset is employed in this experiment with a scaling factor of 20. Given the absence of spatial degradation operator $\mathbf{P}$ and spectral response $\mathbf{R}$ in the real data experiment, we estimate $\mathbf{P}$ and $\mathbf{R}$ with the approach introduced in [34]. The fusion results of the NCALM dataset are displayed in Fig. 12. Still, BSRSV remains stable and reliable performance with the sharpest edges and inconspicuous spectral distortions.

G. Computational Time

The computational time (in seconds) of all methods on the semireal and real datasets are presented in Table VII, which are implemented with MATLAB R2021a on a facility equipped with an Intel Core i9-10900K and 64 GB random access memory.

We can find that the proposed method is less time-consuming than the FuVar and the Gibbs sampling-optimized BSR. Generally speaking, the proposed BSRSV can make a well-balanced tradeoff between the fusion performance and computational complexity.

V. CONCLUSION

In this research, a fully conjugate Bayesian generative model is proposed for HSI–MSI fusion accounting for spectral mismatches between these images caused by different acquisition times. The proposed BSRSV explicitly decomposes the latent HR-HSI into the low- and high-frequency structures separately. The low-frequency structures rely on the up-sampled

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LR-HSI, which can be represented by a spectral dictionary and coefficient matrix. We employ the Gaussian process and Gaussian distribution to facilitate the spectral smoothness and coefficients sparsity respectively. Given the complexity of texture features in the high-frequency components, we employ Gaussian mixture distribution, which has shown sufficient modeling capacity for feature representation. Our BSRSV has several significant characteristics: 1) robustness to variations in spectral signatures between LR-HSI and HR-MSI; 2) robustness to potential spatially varying degeneration operator; 3) less tuning parameters due to adaptive parameter determination mechanism; and 4) the closed-form posterior update for effective and analytical inference. Extensive experiments and comparisons reveal that our BSRSV achieves significant performance gains over state-of-the-art approaches.

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